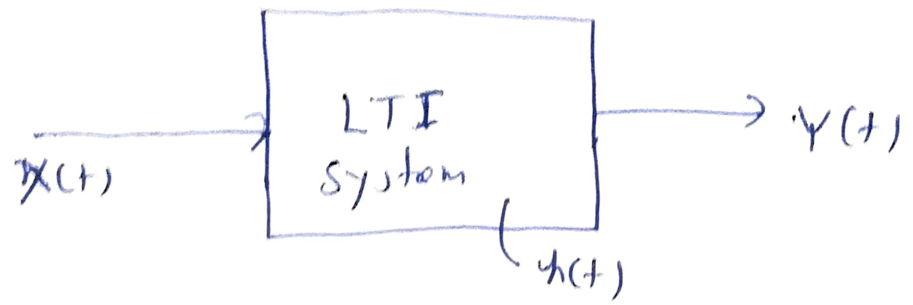
⇒ Transmission of Random process Through Linear system :-



we know that

$$Y(t) = x(t) * h(t)$$

or

$$\Rightarrow \quad y(t) = h(t) * x(t)$$

$$\Rightarrow \quad \boxed{y(t) = \int_{-\infty}^{\infty} h(k)\, x(t-k)\, dk}$$

⇒. Mean and auto correlation of the output:

$$\mu_Y(t) = E[Y(t)]$$

$$\Rightarrow \quad \mu_Y(t) = E\left[\int_{-\infty}^{\infty} h(k)\, x(t-k)\, dk\right]$$

$$\Rightarrow \quad \mu_Y(t) = \int_{-\infty}^{\infty} h(k)\, E[x(t-k)]\, dk$$

$$\Rightarrow \quad \mu_Y(t) = \int_{-\infty}^{\infty} h(k)\, \mu_X(t)\, dk$$

$$\Rightarrow \quad \boxed{\mu_Y(t) = h(t) * \mu_X(t)}$$

②

Auto-correlation:

$$R_{YY}(t_1,t_2) = E[Y(t_1)\,Y(t_2)]$$

$$\Rightarrow R_{YY}(t_1,t_2) = E\left[\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(k_1)\,X(t_1-k_1)\cdot h(k_2)\,X(t_2-k_2)\,dk_1\,dk_2\right]$$

$$\Rightarrow R_{YY}(t_1,t_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(k_1)\,h(k_2)\,E[X(t_1-k_1)\,X(t_2-k_2)]\,dk_1\,dk_2$$

$$\Rightarrow \boxed{R_{YY}(t_1,t_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(k_1)\,h(k_2)\,R_{XX}(t_1-k_1,\,t_2-k_2)\,dk_1\,dk_2}$$

if $X(t)$ is WSS process:

$$R_{YY}(t_1,t_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(k_1)\,h(k_2)\,R_{XX}(t_2-t_1+k_1-k_2)\,dk_1\,dk_2$$

let $\tau = t_2 - t_1$

$$\Rightarrow \boxed{R_{YY}(\tau) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(k_1)\,h(k_2)\,R_{XX}(\tau+k_1-k_2)\,dk_1\,dk_2}$$

③

$$\Rightarrow R_{YY}(\tau) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} R_{XX}(\tau + k_1 - k_2)\cdot h(k_1)\,h(k_2)\,dk_1\,dk_2$$

if $X(t)$ is WSS

$$\Rightarrow \boxed{R_{YY}(\tau) = R_{XX}(\tau) * h(-\tau) * h(\tau)}$$

$\Rightarrow$ Cross-Correlation ~~of~~ Functions of I/p and o/p:-

$$\Rightarrow R_{XY}(t, t+\tau) = E[X(t)\,Y(t+\tau)]$$

$$= E\left[X(t)\int_{-\infty}^{\infty} h(k)\,X(t+\tau-k)\,dk\right]$$

$$= \int_{-\infty}^{\infty} E[X(t)\cdot X(t+\tau-k)]\,h(k)\,dk$$

$$R_{XY}(t, t+\tau) = \int_{-\infty}^{\infty} R_{XX}(\tau-k)\,h(k)\,dk$$

$$\Rightarrow \boxed{R_{XY}(\tau) = R_{XX}(\tau) * h(\tau)}$$

simillaly:

$$\boxed{R_{YX}(\tau) = R_{XX}(\tau) * h(-\tau)}$$

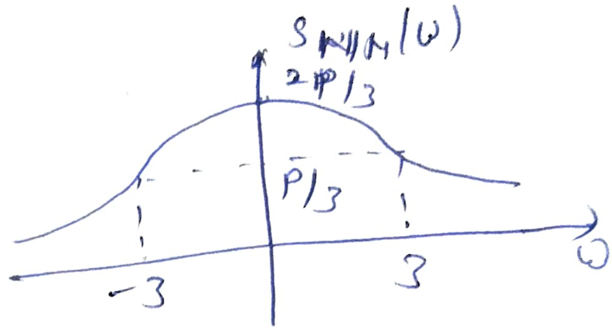
$$\boxed{\begin{aligned} R_{YY}(\tau) &= R_{XY}(\tau) * h(-\tau) \\ R_{YY}(\tau) &= R_{YX}(\tau) * h(\tau) \end{aligned}}$$

Example: A wide-sense stationary noise process $N(t)$ has an ~~at~~ auto-correlation function

$$R_{NN}(\tau) = P e^{-3|\tau|}$$

where P is a constant. ~~the~~ Calculate and ~~draw~~ plot the spectrum of $N(t)$.

Solution:
$$S_{NN}(\omega) = \frac{6P}{9+\omega^2}$$



Example: A random process has the ~~pow~~ autocorrelation function ~~as~~

$$R_{XX}(\tau) = \frac{A^2}{2}\cos(\omega_0 \tau)$$

then, calculate and plot the spectrum of that random process.

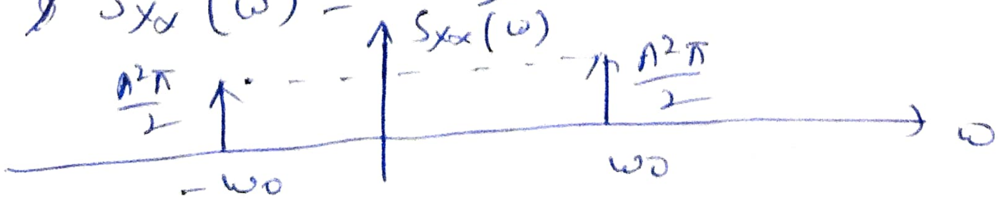
Solution: we know that

$$R_{XX}(\tau) \longleftrightarrow S_{XX}(\omega)$$

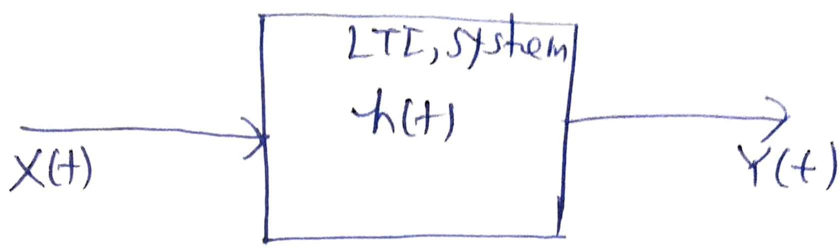
$$\Rightarrow S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau)\, e^{-j\omega\tau}\, d\tau$$

$$\Rightarrow S_{XX}(\omega) = \int_{-\infty}^{\infty} \frac{A^2}{2}\cos\omega_0\tau \; e^{-j\omega\tau}\, d\tau$$

$$\Rightarrow S_{XX}(\omega) = \frac{A^2\pi}{2}\left[\delta(\omega-\omega_0) + \delta(\omega-\omega_0)\right]$$



power density spectrum of Response :-



$\Rightarrow \quad Y(t) = h(t) * X(t)$

since :

$\cancel{S_{YY}(\tau) =}$

$$R_{YY}(\tau) = R_{XX}(\tau) * h(-\tau) * h(\tau)$$

taking F.T. of both side :

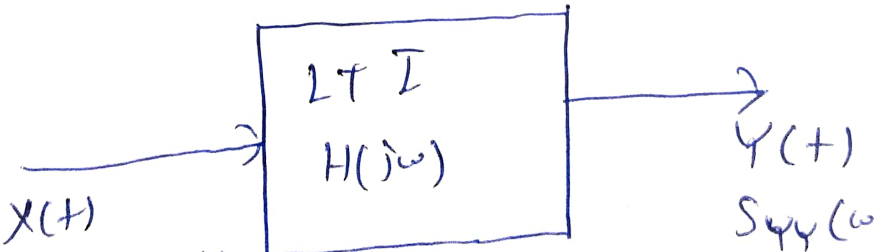
$$F[R_{YY}(\tau)] = F[R_{XX}(\tau) * h(-\tau) * h(\tau)]$$

$$\Rightarrow \quad S_{YY}(\omega) = F[R_{XX}(\tau)] \cdot F[h(-\tau) * h(\tau)]$$

$$\Rightarrow \quad S_{YY}(\omega) = S_{XX}(\omega) \cdot [H(j\omega) \cdot H(-j\omega)]$$

$$\Rightarrow \quad \boxed{S_{YY}(\omega) = S_{XX}(\omega) |H(j\omega)|^2}$$

Example:



$S_{XX}(\omega) = \frac{N_0}{2}$

$S_{YY}(\omega) = ?$

$P_{YY} = ?$

$$H(j\omega) = \frac{1}{\left(1 + j\omega \frac{L}{R}\right)}$$

Solution: we know that: (6)

$$S_{YY}(\omega) = |H(j\omega)|^2 S_{XX}(\omega)$$

$$\Rightarrow S_{YY}(\omega) = \left|\frac{1}{\left(1 + j\frac{\omega L}{R}\right)}\right|^2 \times \frac{N_0}{2}$$

$$\Rightarrow S_{YY}(\omega) = \left[\frac{1}{\sqrt{1+\left(\frac{\omega L}{R}\right)^2}}\right]^2 \times \frac{N_0}{2}$$

$$\Rightarrow S_{YY}(\omega) = \frac{N_0}{2\left(1+\left(\frac{\omega L}{R}\right)^2\right)}$$

$\Rightarrow$ we know that

$$P_{YY} = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{YY}(\omega)\, d\omega$$

$$\Rightarrow P_{YY} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{N_0}{2\left[1+\left(\frac{\omega L}{R}\right)^2\right]}\, d\omega$$

$$\Rightarrow P_{YY} = \frac{N_0}{4\pi}\int_{-\infty}^{\infty} \frac{1}{1+\left(\frac{\omega L}{R}\right)^2}\, d\omega$$

$$\Rightarrow P_{YY} = \frac{N_0}{2\pi}\int_{0}^{\infty} \frac{1}{1+\left(\frac{\omega L}{R}\right)^2}\, d\omega$$

$$\Rightarrow \boxed{P_{YY} = \frac{N_0 R}{4L}}$$

⇒ Cross – power density spectrum of Response:

we know that

$$R_{XY}(\tau) = R_{XX}(\tau) * h(\tau)$$

taking F.T. of both side.

$$F[R_{XY}(\tau)] = F[R_{XX}(\tau) * h(\tau)]$$

$$\Rightarrow \boxed{S_{XY}(\omega) = S_{XX}(\omega) \cdot H(j\omega)}$$

we know that:

$$R_{YX}(\tau) = R_{XX}(\tau) * h(-\tau)$$

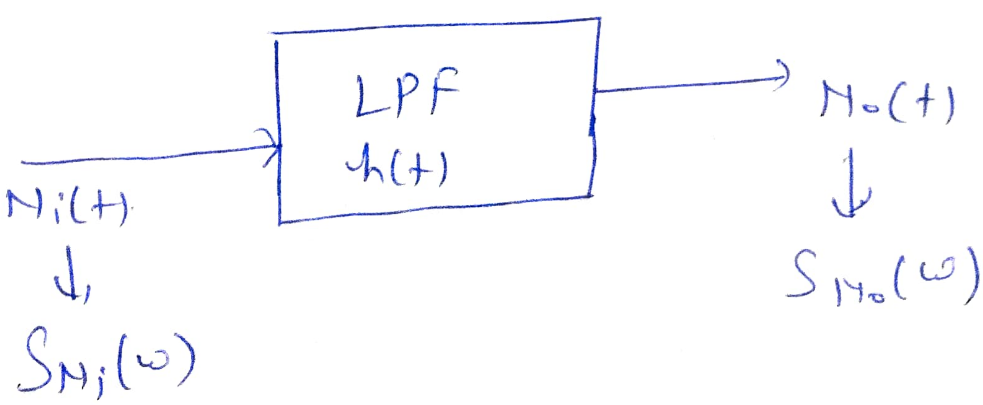
⇒ taking F.T. of both side.

$$F[R_{YX}(\tau)] = F[R_{XX}(\tau) * h(-\tau)]$$

$$\Rightarrow \boxed{S_{YX}(\tau) = S_{XX}(\tau) \cdot H(-j\omega)}$$

(*) white Noise passing through LPF:

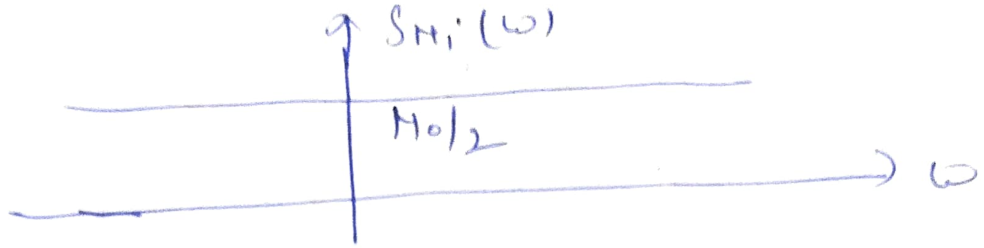
LPF → Low pass Filter.



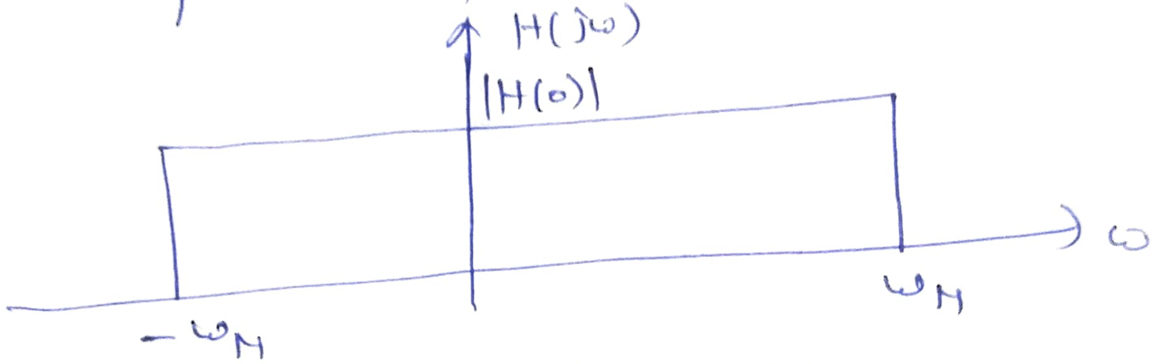
we know that pdf of white noise is given as:

$$S_{N_i}(\omega) = \frac{N_0}{2} \quad -\infty \le \omega < \infty$$

$\Rightarrow$



$\Rightarrow$. Frequency response of LPF is given as.



where $\omega_M \rightarrow$ Noise bandwidth.

$$\Rightarrow \quad H(j\omega) = \begin{cases} |H(0)| & -\omega_M \le \omega \le \omega_M \\ 0 & \text{O.W} \end{cases}$$

$$\Rightarrow \quad (H(j\omega))^2 = \begin{cases} |H(0)|^2 & -\omega_M \le \omega \le \omega_M \\ 0 & \text{O.W} \end{cases}$$

For ideal LPF: $|H(0)| = 1$

$$\Rightarrow \quad |H(j\omega)|^2 = \begin{cases} 1 & -\omega_M \le \omega \le \omega_M \\ 0 & \text{O.W} \end{cases}$$

We know that

$$S_{No}(\omega) = |H(j\omega)|^2 S_{Ni}(\omega)$$

$$\Rightarrow S_{No}(\omega) = 1 \times S_{Ni}(\omega)$$

$$\Rightarrow \boxed{S_{No}(\omega) = \frac{N_o}{2}}$$

$$P_{No} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{No}(\omega)\, d\omega$$

$$\Rightarrow P_{No} = \frac{1}{2\pi} \int_{-\omega_M}^{\omega_M} \frac{N_o}{2}\, d\omega$$

$$= \frac{N_o}{2\pi} \int_{0}^{\omega_M} d\omega$$

$$\boxed{P_{No} = \frac{N_o}{2\pi} \times \omega_M}$$

$$\Rightarrow \boxed{\text{Power} = \frac{N_o}{2\pi} \times B.\omega}$$

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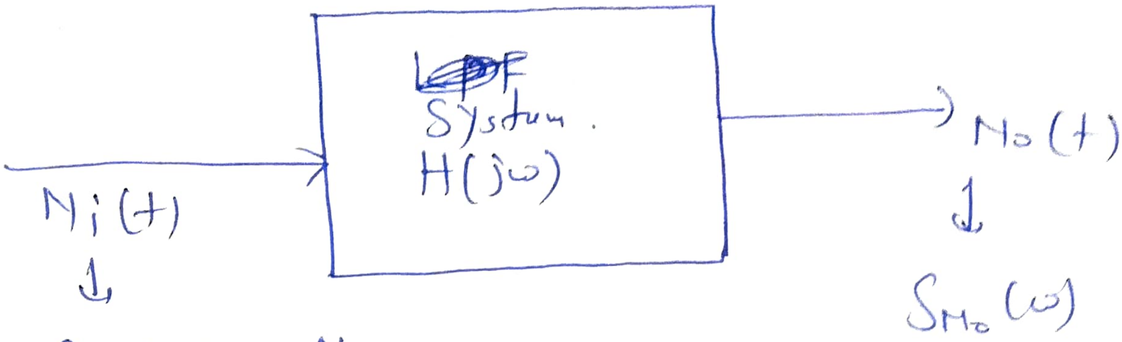
Noise Bandwidth:



$$S_{Ni}(\omega) = \frac{N_o}{2}$$

we know that o/p power of Noise is given by:

$$P_{No} = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{N_o}{2}\right)|H(j\omega)|^2\, d\omega \quad \text{—} (I)$$

For ideal LPF:

$$|H(j\omega)|^2 = \begin{cases} |H(0)|^2 & -\omega_H \le \omega \le \omega_H \\ 0 & \text{o.w} \end{cases}$$

$\Rightarrow$ From eqn. (I)

$$P_{No} = \frac{1}{2\pi}\int_{-\infty}^{\infty}|H(0)|^2 \times \frac{N_o}{2}\, d\omega$$

$$\Rightarrow P_{No} = \frac{2}{2\pi}\int_{0}^{\omega_H}|H(0)|^2 \times \frac{N_o}{2}\, d\omega$$

$$\Rightarrow P_{No} = \frac{N_o}{2\pi}\int_{0}^{\omega_H}|H(0)|^2\, d\omega \quad \text{—} (II)$$

(10) (11)

From eqn. (1) & (11)

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{N_0}{2}|H(j\omega)|^2 d\omega = \frac{N_0}{2\pi}\int_0^{\omega_H}|H(0)|^2 d\omega$$

$$\Rightarrow \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{N_0}{2}|H(j\omega)|^2 d\omega = \frac{N_0|H(0)|^2}{2\pi}\times\omega_H$$

$$\Rightarrow \frac{N_0}{2\pi}\int_0^{\infty}|H(j\omega)|^2 d\omega = \frac{N_0|H(0)|^2}{2\pi}\times\omega_H$$

$$\Rightarrow \boxed{\omega_H = \frac{\int_0^{\infty}|H(j\omega)|^2 d\omega}{|H(0)|^2}}$$